Homelab architecture

Physical hosts, service placement and the shared platform.

People & devices
Browser · phone · workstation
Home LAN or remote Tailscale

Two access paths
Public: Internet → router → Caddy → selected services
Private: LAN / Tailscale → Caddy or native LAN protocols
Tailscale routing: px10 / px20 active; px30 powered off.

Cloudflare
DNS-only records · DNS-01 TLS
Private defaults; public exceptions
DNS is not a traffic tunnel.

slate · external GCP VM
Caddy · Uptime Kuma · ntfy
Independent monitoring & alerts
Checks home over Tailscale

LAN fabric / Netgear GS110TPv3 — Ethernet: 1 Gb/s full duplex verified at px10, px20 and px30 — Wi-Fi → bulb & AC

px10 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · NVMe ZFS / SATA ext4

- **dev · VM 100**
  Codex · Claude Code · Hermes · OpenCode
- **observability · LXC 104**
  Beszel · Grafana · VictoriaMetrics
- **store · LXC 109 · stopped / shares retired**
  Samba stopped; old AV/BACKUP layout retired
- **dns · LXC 102**
  AdGuard Home → Unbound recursion
- **auth · LXC 103**
  Pocket ID / OIDC · Tinyauth / forward-auth
- **s3 · LXC 105 · stopped / root erased**
  SATA root erased; cold file backup verified
- **k8s-201 · VM 201 · stopped / data erased**
  NVMe OS retained; former SATA data disk erased

px20 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · NVMe ZFS / SATA ext4

- **ctr · VM 204 · stopped / restore data / Docker + Portainer**
  Application data erased; restore backup before start
  Personal Immich · Karakeep · OpenCloud · Paperless
- **orva · VM 106**
  Serverless functions · Quick Share
- **proxy · LXC 101**
  Caddy · HTTPS routing / TLS
- **haos · VM 107**
  Home Assistant · Mosquitto · local device control
- **k8s-202 · VM 202 · stopped / data erased**
  NVMe OS retained; former SATA data disk erased

px30 · powered off — Dell OptiPlex 5060

i5-9400T / 6 cores · 32 GB RAM · ZFS OS + data pools

- **k8s-203 · VM 203 · host unreachable / NotReady**
  Talos · control plane + workloads
  4 vCPU · up to 16 GiB RAM · 500 GiB data

Placement is a snapshot; managed guests can relocate.

Kubernetes stopped for retirement; backups verified

Talos management: Omni on ctr. Kubernetes desired state: GitHub → Argo CD.

Networking & operations

Cilium CNI · L2 LoadBalancer services · CoreDNS
Argo CD · Headlamp · Homepage · metrics-server
vmagent · kube-state-metrics · node metrics

Persistent storage

Cluster stopped; px10/px20 data disks erased
Logical application backups verified on external SSD
px30 local disk remains inaccessible and untouched

Supabase

Offline; authoritative SQL and application files saved
Studio · Auth · REST · Realtime · Edge Functions
Storage / imgproxy · Kong · pooler · logs / analytics

Storage boundaries

NVMe ZFS roots retained; SATA data erased
4 TB backup; 256 GB USB overwrite in progress
CTR / S3 / Kubernetes require restoration from backup

Availability boundaries

NVMe services retained; data services stopped
Dev, DNS, identity and proxy remain on NVMe
Restore CTR / S3 / Kubernetes from the backup

Smart home

Home Assistant → Wipro bulb / Tuya LAN
Home Assistant → Samsung AC / local DTLS
Frigate offline; recordings retained in backup

Switch and host-port map verified; router/AP models and remaining physical cabling are unverified.

How traffic and data move

The important connections, without drawing every packet or replica.

Web access & identity

Client
LAN / Tailscale for private apps
Internet for selected public apps

→

Caddy · proxy LXC
HTTPS termination → backend routes
Source restrictions per service

→

Application
Native login / API credentials
No blanket public or SSO bypass

Pocket ID + Tinyauth · auth LXC
Native OIDC where supported
Caddy forward-auth for selected browser routes

Name lookup: AdGuard → Unbound

Identity is a policy dependency; API/MCP routes retain their own authentication.

GitOps & cluster operations

GitHub · Homelab repository
Declarative non-secret configuration
Reviewed changes → desired state

⇢

Argo CD · Kubernetes
Reconciles tracked applications
Signed public webhook; private UI

⇢

Talos / Kubernetes
Cilium network · Longhorn volumes
k8s-201 / 202 active; k8s-203 stopped

⇠

Omni · ctr
Manages the three Talos machines
Runs on ctr, not inside Kubernetes

Longhorn: storage-level replicas. CloudNativePG: separate PostgreSQL streaming replicas and failover.

Agents & shared memory

dev · four agents
Codex · Claude Code
Hermes · OpenCode

→

OpenViking · ctr
Shared account-scoped memory
Native plugins / provider and MCP

→

Ollama · ctr
Local embedding service
Not the agents' primary chat model

Tools & integrations
Orva · Nivra · Supabase · browser / developer MCPs
Hermes → Karakeep CLI / Home Assistant API

Hermes: Go / DeepSeek + Codex fallback. OpenCode: Claude Agent SDK. Telegram / iMessage → Hermes.

Media & object storage

Media path
qBittorrent / Motrix → Store AV share → Jellyfin
Encrypted SMB3 directly over the LAN; not through Caddy.
Media stays on Store's USB SSD, not ctr's OS disk.

Temporary sharing
Quick Share CLI → Orva function → Cairn S3
Short Orva share URL → function proxy → object download.
Cairn is a separate native service on the s3 LXC.

Container images
Docker client → Registry → RustFS S3
Registry runs on ctr; image layers live in RustFS.
Cairn and RustFS share one LXC / host failure domain.

Proxmox nodes use Store BACKUP / ISO over SMB directly. App databases remain on their own local or block volumes.

Home automation & visibility

Hermes / user
Home Assistant API / app
Existing authenticated access

→

Home Assistant · haos
Wipro bulb → Tuya LAN
Samsung AC → local DTLS / CoAP

←

Frigate · ctr / Intel iGPU
Camera ingest / recording
RTSP restream + MQTT → HA

Metrics · observability LXC
Exporters + Kubernetes vmagent → VictoriaMetrics
Grafana / HA node views; Beszel / Kuma monitoring

Local device protocols avoid cloud commands; a full WAN-outage acceptance test remains outstanding.

Solid arrows: application / data path → Dashed arrows: configuration / dependency ⇢ Not a live health map; see the inventory notes for verification limits.

Schemas are simplified; replicated storage and HA are not independent backups.